

1-A:

- Since the project has a **social cost of \$100m**, and a **private cost of \$40 m**, it can be said that EC is "**60**"($100-40$)
- And since, the net social value is "0", this means that $SB-SC=0$. Since, we are given **social cost of \$100m**, the **social benefit must also be \$100m**, for the net social value to be zero.
- The social benefit is \$100m, and the external benefit is \$20m, this means that the **private benefit must be \$80**.

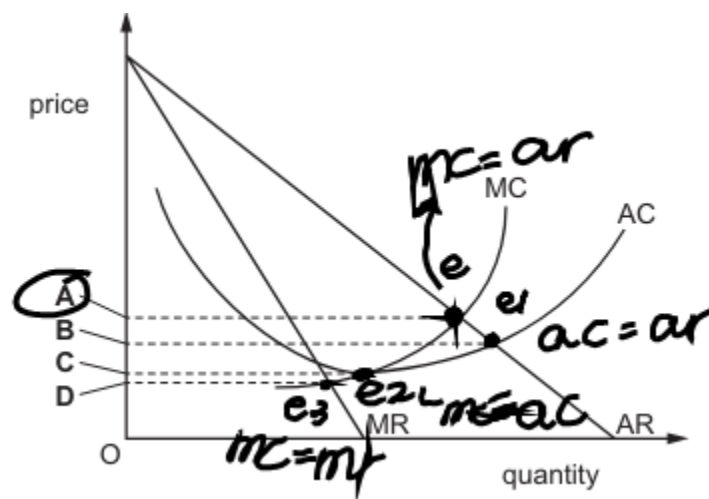
A – This is correct because **external cost i.e. 60** is greater than **external benefit of 20**

B – **Private cost(\$40m)** is not greater than **external cost(\$60m)**

C – **Private cost(\$40m)** is not greater than **private benefit(\$80m)**

D – **Social cost(of \$100m)** is equal to **social benefit (of \$100m)**

2-A:



A - It can be seen in the diagram above that price level A represents allocative efficiency because that is when $MC=AR$.

B – Price level B represents that normal profits are made because that is when $AC=AR$.

C – Price level C represents productive efficiency because $MC=AC$ there. When MC intersects with AC, this represents that average cost has been minimized.

D- Price level D represents profit maximizing level of output because $MC=MR$.

3-B:

A government is considering improving the rail links in its country. It also has to choose one of four high-speed routes.

The benefits and costs of each route are shown below.

Which route should be chosen?

SB	private benefits \$bn	+	external benefits \$bn	private costs \$bn	+	external costs \$bn	SC	Net SB
34	14	+	20	0.4	+	0.1	0.5	33.5
38	16	+	22	1.6	+	0.4	2.0	36
30	18	+	12	0.2	+	1	1.2	28.8
38	20	+	18	2	+	4	6	32

$$* \text{Net SB} = \text{SB} - \text{SC}$$

highest
net
social
benefit
↓
Option B

-> Since the project that yields the highest net social benefit will be selected, option B becomes correct.

4-D:

The table shows the total utility gained by a consumer as they consume more apples.

apples consumed	total utility	MU
0	0	0
1	10	10
2	18	8
3	24	6
4	28	4
5	30	2
6	30	0

$MU=0$
 $TU=MAX$

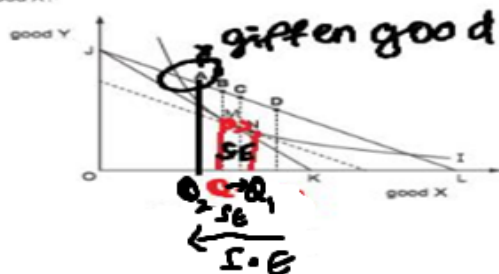
At which level of consumption would the marginal utility be zero?

A 1 B 4 C 5 D 6

It can be seen in the diagram above that when 6 apples are consumed, TU is maximized at 30 utils, and so, MU is zero at that point. This is why key D becomes correct.

5-A:

In the indifference curve diagram point M is the consumer's initial equilibrium. JK and JL are budget lines and MN is the substitution effect of a fall in the price of good X.
If good X is a Giffen good, which point will be the consumer's new equilibrium point after the fall in the price of good X?



Handwritten notes:
 $SE \rightarrow$ extension in qty demanded from $Q_1 - Q_2$
 $IE \rightarrow$ contraction in qty demanded from $Q_1 - Q_2$
 IE > SE

Since income effect outweighs the substitution effect, the good will be a "giffen good" if the final equilibrium is A. In case of giffen goods, there is a positive relationship between price and qty demanded. If for example, the price of a loaf of bread falls, people will contract some qty of loaves to purchase some meat which they can now afford, in order to feel richer. This is why, key A, will be correct.

6-A:

Selina has an income of \$100 and buys food and drink. The price of a unit of food is \$2 while the price of a unit of drink is \$1.

Which combination of spending would suggest that Selina wants to save some of her income?

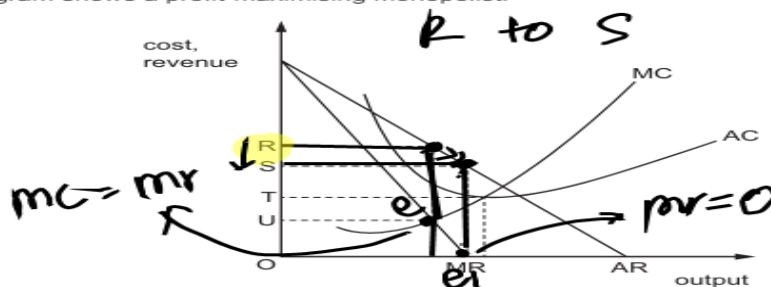
	food (units)	drink (units)	Total
A	20 $\times 2 = 40$	50 $\times 1 = 50$	90
B	30 $\times 2 = 60$	40 $\times 1 = 40$	100
C	40 $\times 2 = 80$	20 $\times 1 = 20$	100
D	50 $\times 2 = 100$	10 $\times 1 = 10$	110

Since a total of \$90 has been spent in option A, this represents that Selina may have wished to save \$10 of her income. This is why, key A, is correct.

7-A:

When the objective changes from profit maximization (that is when $MC=MR$) to revenue maximization (that is when $MR=0$), the price level falls from R to S.

7 The diagram shows a profit-maximising monopolist.



What would be the change in price if this monopolist changed from profit maximisation to revenue maximisation? $mc=mr$

8-B:

The correct answer is B) profit satisficing.

Profit satisficing is a goal where a firm aims to earn a minimum acceptable level of profit, rather than maximizing profit which is enough to keep shareholders happy.

A) Profit maximization ($MC=MR$) aims to maximize profits, not just reach a minimum level.

C) Revenue maximization ($MR=0$) focuses on maximizing total revenue, not profit.

D) Sales maximization ($AC=AR$) focuses on maximizing sales volume, not profit

9-D:

A. This option is incorrect because decreasing long-run average costs would actually allow the monopolist to increase its profits, not reduce them to normal levels. With decreasing average costs, the firm can produce more at a lower cost, giving it a greater incentive to expand output and increase profits.

B. In case of a public company, there is more likely to be a principal agent problem or in other words, conflict of interest between shareholders and managers. And it is most likely that due to information asymmetry, the objective of revenue maximization (where $MR=0$) is kept.

C. The ownership structure of the firm, whether it is owned by a few financial institutions or not, does not directly determine its profit levels in the long run.

D. This option is the correct answer. A highly contestable market means that there are no significant barriers to entry, and potential competitors can easily enter the market if they see an opportunity for profit. This competition prevents the monopolist from earning more than normal profit in the long run.

10-B:

A – This statement is correct because firm X will not choose to produce at all. The reason behind it not operating in the short-run is that the Average revenue or the price does not cover Average variable cost, and also, it will not operate in the long run because the AR does not cover ATC.

B – This statement is not correct **which is why this option becomes right**. Firm Y is likely to operate in the short-run and not in the long-run because the Average revenue does cover Average variable cost but it does not cover Average total cost.

C – This statement is correct because Firm Y is likely to operate in the short-run and not in the long-run because the Average revenue does cover Average variable cost but it does not cover Average total cost.

D – This is correct too as firm Z will operate in both, the short run and the long run because the Average revenue covers AVC which is essential for survival in the short-run, and also covers ATC which is essential for survival in the long-run.

11-B:

A: Decreasing marginal costs would have occurred when marginal product of labor would have risen.

B: All inputs changing means that “the capital inputs” are also changing, and none of them is fixed, which represents that we are operating in the long-run. Since inputs are greater than outputs, this means that the long run average costs are rising, which represents a situation of “dis-economies of scale” or “decreasing returns to scale”. This is why, this is correct

C: Economies of scale would have occurred if output was greater than inputs.

D: All inputs changing means that “the capital inputs” are also changing, and none of them is fixed, which represents that we are operating in the long-run. The law of diminishing returns would have occurred in the short-run, when at-least one input (capital mainly) was fixed. This is why, this is incorrect.

12-B: → Profit represents the “income” of the factor “enterprise”

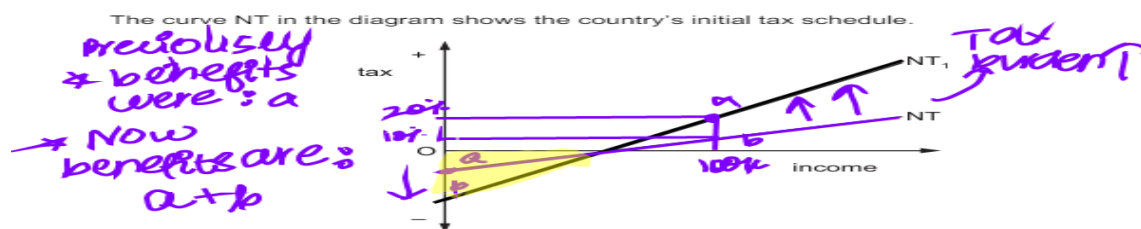
A – Excess profit represents the “economic rent” earned by the factor enterprise.

B – Normal profit is an economic term that describes a situation in which a company’s total costs are equal to total revenues. It represents “zero-economic profits, and this is the minimum amount that the entrepreneur needs to stay in the industry since it represents the opportunity cost of resources invested in the enterprise. As a result of which, this represents the “transfer earnings”. This is why, key B, becomes correct.

C – Return on capital employed represents how much each \$1 of resources invested in the venture earns for the business in terms of profits. It does not represent “transfer earnings” for the factor: enterprise.

D – “Start-up costs” refer to the initial expenses incurred when setting up a new business, project, or venture. These costs are associated with preparing and launching the business or project and include various expenditures required to get operations off the ground and running. This is incorrect because costs do not represent earnings.

13-D:



A change in the tax rate causes the schedule to shift to NT₁.

How will this affect work incentives and the after-tax distribution of income?

	work incentives	distribution of income
A	strengthen	more equal
B	strengthen	less equal
C	weaken	less equal
<u>D</u>	<u>weaken</u>	<u>more equal</u>

In the diagram, it can be seen that the income tax system has been made more progressive and also, the amount of means tested benefits provided to individuals have also increased by the region b. This indicates that **the distribution of income has become more equal**.

Similarly, since the rate of taxation has risen, it can be said that, work incentives have weakened. Individuals would be less likely to work now because they know that a greater proportion of additional incomes earned will go in tax.

14-C:

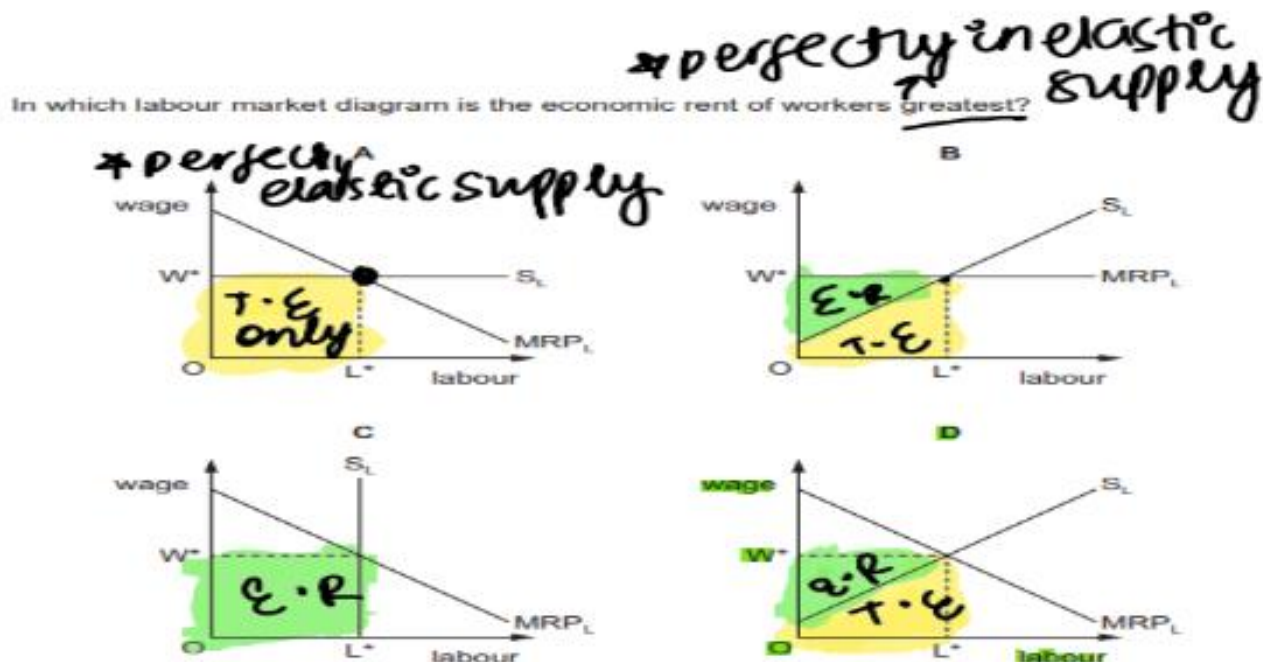
→ Transfer earnings are the minimum income a worker needs in order to supply their labor. Economic rent is the extra income a worker receives – above the minimum level they need in order to work.

→ The area below the supply curve represents “transfer earnings”. The area above the supply curve and below the wage rate represents the “economic rent”

→ In a **perfectly inelastic supply of labor**, the quantity of a specific type of labor remains constant regardless of changes in wage rates. This is often the case for highly specialized or unique skills where there are only a limited number of individuals who possess those skills.

Due to the scarcity of workers with unique skills, employers offer wages that are higher than the minimum required to attract and retain them. This excess payment above what is needed to prevent them from leaving for other opportunities is considered “economic rent”

→ Therefore, the wage of such workers only comprises of “**economic rent**”



15-A:

The correct answer is A. Government failure may result from policies that have unintended side effects.

Government intervention can sometimes lead to unintended consequences, which may even be worse than the problem they were trying to solve. This is known as government failure. Policies may have unintended side effects, such as creating disincentives or distortions in the market, which can ultimately reduce efficiency and welfare.

B. This is incorrect because governments can identify inefficiencies in the market, such as market failure associated with negative externalities.

C. The presence of government failure does not necessarily mean that the market would have done better, and vice versa.

D. This is also incorrect because government intervention is not always the solution to market failure. Government policies may sometimes exacerbate the problem or create new inefficiencies, leading to government failure.

16-D:

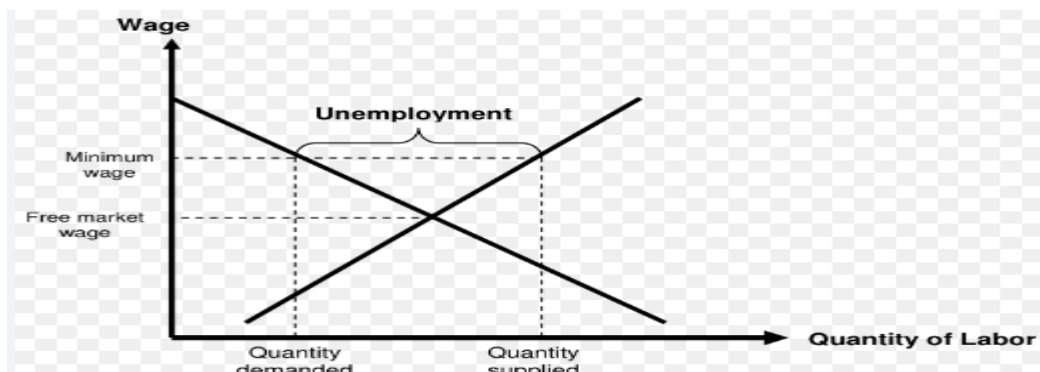
A – There is no guarantee that ‘**all**’ the workers will receive the new guaranteed wage, because some will be paid higher than the minimum wage.

B – This represents an extension in quantity supplied of labor after the minimum wage legislation is enforced, but the extent to which there is an extension depends upon “price-elasticity of supply”

C -Although there is an extension in the supply of labor, there will be **a fall in the level of employment after the minimum wage legislation is enforced**. This is because while quantity supplied extends, there is a contraction in quantity demanded (equilibrium qty) of labor.

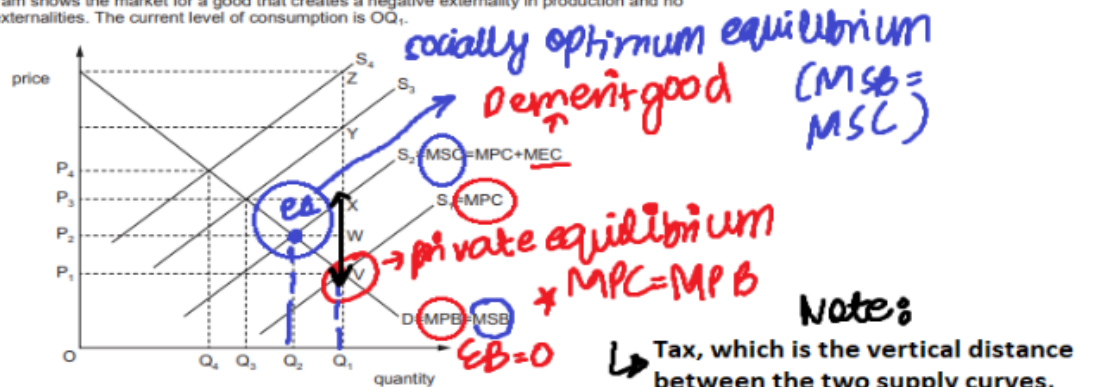
D – After the minimum wage legislation is enforced “**above the equilibrium**”, there will be a contraction in qty demanded and an extension in qty supplied of labor, representing an existence of “surplus of labor in the labor market” representing “unemployment”. This can be seen in the diagram below.

Therefore, this option becomes correct.



17-B:

The diagram shows the market for a good that creates a negative externality in production and no positive externalities. The current level of consumption is OQ_1 .



The government decides to tax producers of the good.

Which size of tax on producers would result in a socially efficient allocation of resources?

- A VW **B VX** C VY D VZ
Tax = size of external cost

This explains why, key B, is correct.

18-B:

Since the question says that there has not been a potential growth, this represents that the production possibility curve will not shift rightwards; this is why, options C and D become incorrect.

Actual growth occurs when actual (point inside the PPC shown through X) moves closer to the potential that is shown through point B. This explains why key B will be correct.

19-A:

19 The information in the table is taken from a country's national income accounts

	\$ million
consumer expenditure	250
investment expenditure	100
government expenditure	150
exports	100
imports	150
taxes	80
subsidies	40

What is the value of national income at factor cost in \$ million?

- A 410**
 B 450
 C 500
 D 550

** National income at market values*

$$* C + I + G + (X - M)$$

$$* 250 + 100 + 150 + (100 - 150)$$

$$* 450$$

Formula:

*gap at factor cost
 + taxes
 - subsidies
 ⇒ MV*

$$\begin{array}{r} x \Rightarrow 410 \\ 80 \\ (40) \\ \hline 450 \end{array}$$

20-A:

A – Pensions paid to retired people are transfer payments that are payments made to individuals without expectation of goods or services or any contribution to the gdp in exchange. This is why, they are not included in measuring real GDP.

B , C and D- Profits made by firms, rent paid to landlords and wages paid to nurses are all economic returns arising as a result of economic activities. This is why, they will be included when measuring real GDP.

21-C:

A , D: **Differences in government spending and value of exports have already been accounted in real gnp figures.**

B: Since we have data of real gnp per head, it can be said that adjustment for inflation has already been made.

C: The hidden economy, also known as the informal or shadow economy, refers to economic activities that are not reported or taxed. This can include activities like black market transactions, unpaid household work, or illegal activities. Adjusting for the size of the hidden economy can help ensure that the comparison is made on a more comprehensive basis. This is correct because it prevents the gnp figures from being understated.

22-A:

The Kuznets curve has an inverted U-shape, with income inequality increasing as a country moves from a low-income to a middle-income economy, and then decreasing as it moves from a middle-income to a high-income economy. This curve represents the idea that economic growth is initially accompanied by increasing inequality, but eventually leads to a more equal distribution of income. **This explains why option A is correct.**

Here's why the other options are incorrect:

B. Human Development Index (HDI): The HDI is a measure of a country's well-being, taking into account factors like health, education, and income.

C. Multidimensional Poverty Index (MPI): Like HDI, MPI is not directly related to income inequality.

D. Lorenz curve: The Lorenz curve is a graphical representation of income inequality, showing the proportion of total income earned by different segments of the population. While it's related to income inequality, it's not the same as the Kuznets curve, which shows how inequality changes over time.

23-A:

$$23 \Rightarrow \text{multiplier} \Rightarrow \frac{1}{\text{mpb} + \text{mps} + \text{mpm}} \Rightarrow \frac{1}{0.3 + 0.2 + 0.3} \Rightarrow 1.25$$

24-B:

24 The information in the table is taken from a country's national income accounts.

	\$ million
national income	600
consumer spending	400 ✓
investment spending	80 ✓
government spending on goods and services	100 ✓
exports	140 ✓

What is the value of imports in \$ million?

- A 100 **B 120** C 140 D 240

$$\begin{aligned} * C + I + G + (X - M) &= 600 \\ 400 + 80 + 100 + (140 - X) &= 600 \\ 580 + 140 - X &= 600 \\ 720 - X &= 600 \\ 720 - 600 &= X \\ \boxed{X &= 120} \end{aligned}$$

25-B:

Quantitative easing is a monetary policy tool used by central banks to inject liquidity into the economy. In QE, the central bank buys back government securities from the non-bank private sector, as a result of which, the money supply increases. This injection of liquidity aims to stimulate economic activity, lower interest rates, and increase lending.

This is a part of expansionary monetary policy.

Here's why the other options are incorrect:

A. Liquidity preference: This term refers to the preference of individuals and businesses to hold money, rather than investing in illiquid assets. This also **represents money demand**.

C. Supply-side policy: This term refers to economic policies aimed at either improving the productive capacity of an economy or are known to boost the short run aggregate supply of the economy.

D. The transmission mechanism: This assesses how changes in money supply impact interest rates, which further impact investment decisions and subsequently Aggregate demand.

26-C:

- ➔ In a fast-growing emerging economy, GDP per head rises due to increasing productivity, technological advancements, and investments in human capital.
- ➔ As people become more affluent, they tend to have fewer children and invest more in their education and well-being. This is why, birth rates fall.
- ➔ In a fast-growing emerging economy, the household savings ratio rises as people's incomes increase and they become more financially secure. With higher incomes, individuals and households are able to save more for the future.

27-B:

A – An increase in retraining facilities will improve the occupational mobility of labor because of which, structural unemployment will reduce.

B – A significant increase in indirect taxes will increase costs of production of producers in an economy, as a result of which, Aggregate supply will reduce. This not only brings about a shortage in an economy which results in an upward pressure prices, representing cost push inflationary pressures, but also translates into a fall in real incomes, which illustrate a fall in incomes, output and employment.

C – Imposition of import controls will result in import substitution and when because of this domestic consumption increases, Aggregate demand will increase. And this further leads to a rise in production and demand for labor, because of which, cyclical unemployment falls.

D – Increase in government's fiscal deficit represents expansionary fiscal policy which is aimed to increase Aggregate demand and subsequently reduce cyclical unemployment.

28-C:

A) This option would be counterproductive in reducing cyclical unemployment. Decreasing government expenditure would lead to a reduction in aggregate demand, causing businesses to produce less and hire fewer workers. Additionally, increasing income tax would reduce disposable income, leading to lower consumption and a reduction in Aggregate demand, thereby exacerbating unemployment.

B) A budget deficit reduction could lead to a decrease in government expenditure or an increase in taxes, which could further reduce aggregate demand and worsen unemployment.

C) This option is the most likely to be effective in reducing cyclical unemployment. Increasing government expenditure would boost aggregate demand, leading to higher production and employment. Decreasing income tax would increase disposable income, encouraging consumption which results in an increase in Aggregate demand, thereby stimulating economic growth and job creation.

D) :Increasing the budget surplus would likely lead to a reduction in government expenditure and/or an increase in taxes, which would reduce aggregate demand and worsen cyclical unemployment.

29-D:

The Phillips curve represents the historical inverse relationship and the **trade-off** between unemployment and inflation. As the economy adopts an expansionary monetary policy to boost employment, the curve suggests that:

- Lower unemployment leads to higher inflation.

In this scenario, the accelerating rise in the consumer price index (CPI) indicates increasing inflation, which is consistent with the Phillips curve.

Here's why the other options are incorrect:

A) Kuznets curve: Represents the relationship between economic growth and income inequality over time.

B) Laffer curve: Represents the relationship between tax rates and government revenue.

C) Lorenz curve: Represents income distribution and wealth inequality.

30-C:

The correct answer is C) increase general tariffs on imports.

To protect jobs, the government of country X is likely to implement policies that limit competition from foreign workers and goods. Increasing general tariffs on imports would make imported goods more expensive, thereby reducing imports.

And when because of this import substitution, domestic demand rises, there is an increase in production and demand for labor which results in a **fall in demand-deficient unemployment**.

A) Decreasing restrictions on immigrant labor would likely increase competition for domestic workers and potentially lead to job losses.

B) Decreasing subsidies to domestic producers would reduce their competitiveness in the international market and when export demand falls, the export oriented firms are most likely to contract their scale of production, and lay off workers.

D) Removing quotas on imports might lead to an increase in imports. And when because of this domestic consumption and subsequently Aggregate demand falls, production and demand for labor decreases that results in the economy experiencing a rise in cyclical unemployment.

